Backtest — the comparisons that matter

Five funds from **backtest_data/results_fixture/** | cutoffs 2022-05-31 to 2024-06-30 | 268 validated predictions | SIP Rs 10,000/month

Before any number below: this run cannot tell you if the model picks good funds.

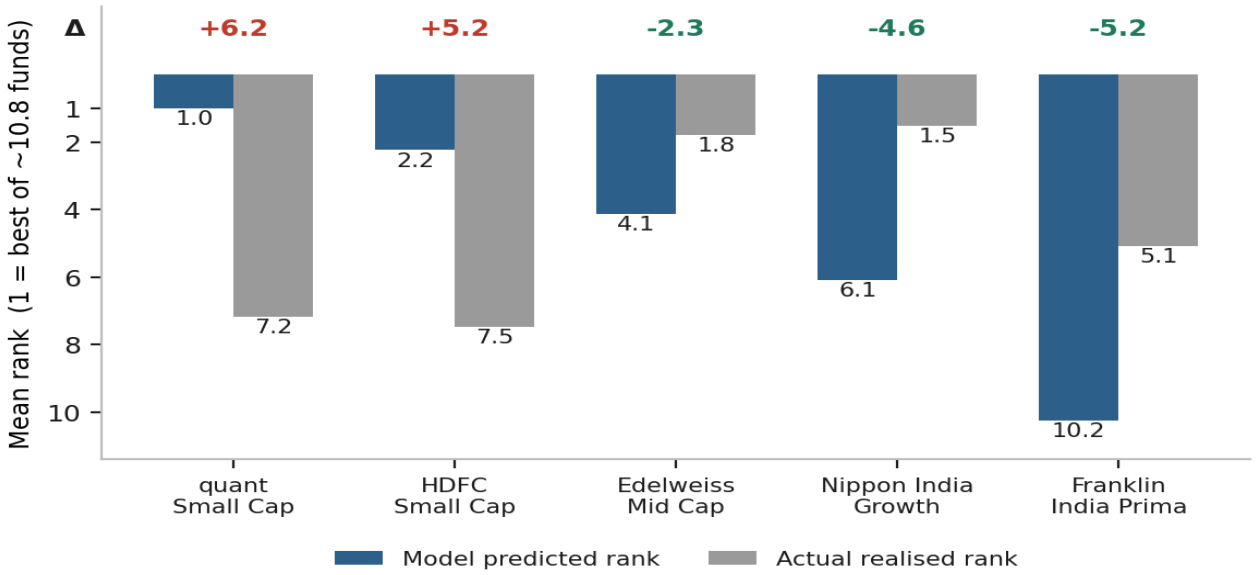
The fixture has **2.04 independent periods** — 25 cutoffs of overlapping 2-year windows contain about two genuinely independent observations. **accuracy_summary.json** marks every rank metric *"not interpretable — too few independent periods"*, and that verdict is correct.

The window (2022-05 to 2024-06) is also a **single regime**: a small-cap-led rally followed by rotation. That one fact explains most of what follows. What the run *is* good for: confirming the machinery works, and exposing one specific failure mode worth testing on real 14-year history.

The five funds at a glance

Code	Fund	n	Pred rank	Actual rank	Δ	Pred WM	Actual WM	WM error	Above p10	Mean FQS
100177	quant Small Cap	24	1.00	7.17	+6.17	1.626	1.186	-0.440	0.25	82.6
130502	HDFC Small Cap	25	2.24	7.48	+5.24	1.427	1.155	-0.272	0.28	78.9
140225	Edelweiss Mid Cap	24	4.12	1.79	-2.33	1.363	1.260	-0.102	0.79	76.3
105503	Nippon India Growth	25	6.08	1.52	-4.56	1.288	1.269	-0.019	0.80	70.0
104481	Franklin India Prima	25	10.24	5.08	-5.16	1.204	1.184	-0.020	0.96	61.8

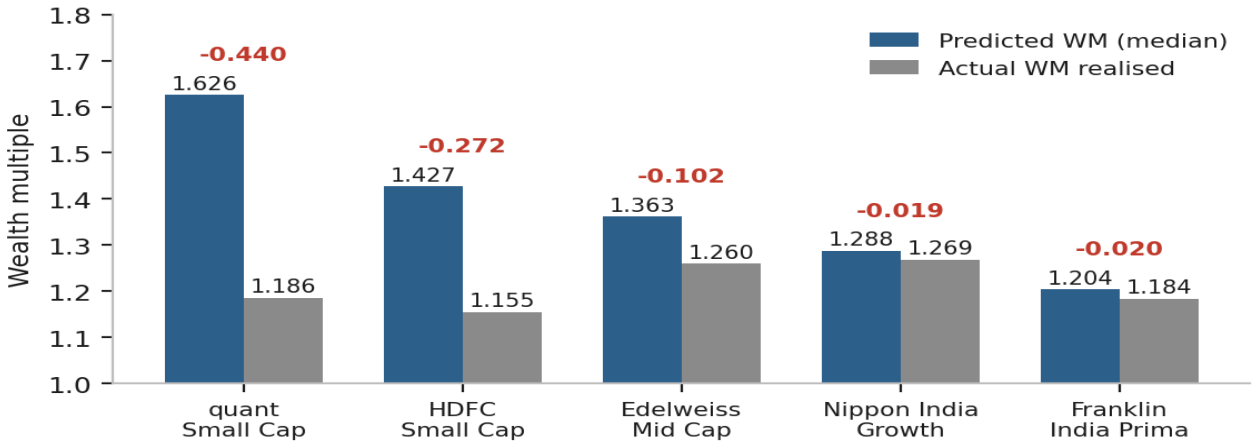
Rank 1 = best of ~10.8 funds. WM = wealth multiple. *Above p10* = share of cutoffs where realised WM cleared the model's own p10 floor (should be ~0.90).



quant Small Cap was ranked #1 at all 24 validated cutoffs without exception, and finished 7.2 of ~10.8 — worst single cutoff missed by 10 places. Nippon India Growth landed in the actual top decile 16 of 25 cutoffs; the model never flagged it top-decile once.

Comparison 2 — predicted wealth vs realised wealth

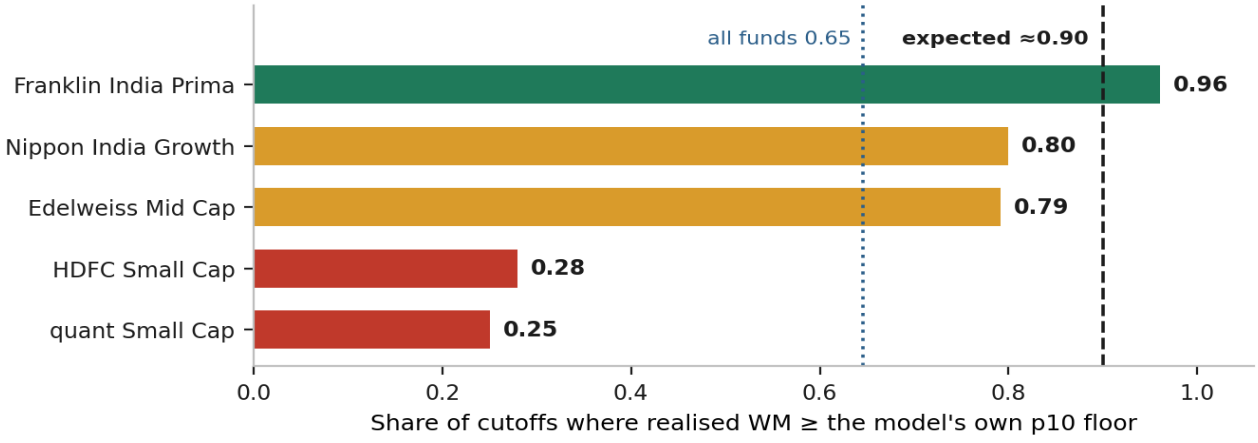
The most trustworthy finding here, because it is arithmetic on realised outcomes rather than inference about ranking skill.



Across all 268 validated predictions	Value
Mean predicted WM	1.3238
Mean actual WM	1.1727
Mean error	-0.1511
Median error	-0.1321
Mean absolute percentage error	12.4%

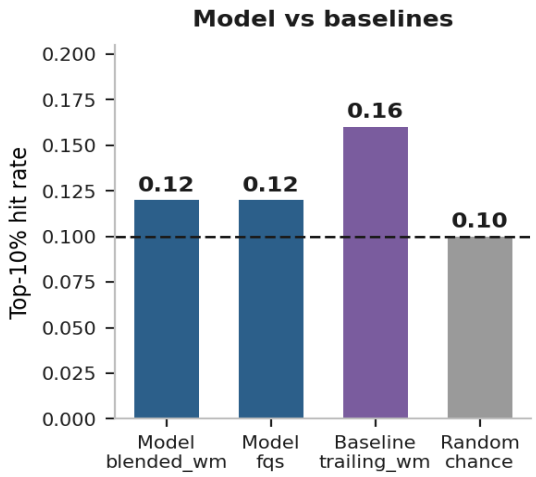
All five funds miss in the same direction. A consistent sign across every fund is a bias, not sampling error — and unlike the ranking results, this one survives the two-independent-periods objection. On a Rs 10,000/month SIP the quant Small Cap gap is a Rs 105,543 mean shortfall; HDFC Small Cap Rs 65,246.

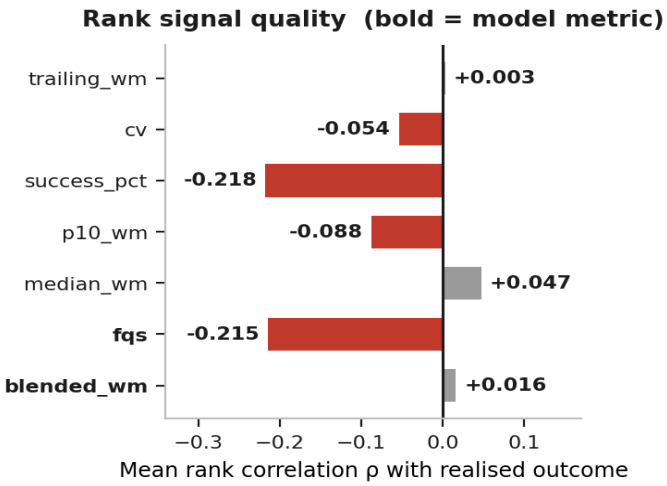
Comparison 3 — is p10 behaving like a downside case?



p10 is presented to users as a downside case, so realised outcomes should clear it about 90% of the time. Observed: 64.5%. The report's own note flags this as miscalibration. It concentrates badly — 0.25 and 0.28 for the two small caps against 0.96 for Franklin Prima. The floor fails precisely where a user would lean on it hardest.

Comparison 4 — model vs baselines





Selection method	Picked actual winner	Top-10% hit rate	Mean percentile of picks
Model — blended_wm (the app's rank metric)	0.12	0.12	0.439
Model — fqs (headline score)	0.12	0.12	0.490
Baseline — trailing_wm (one recent window)	**0.16**	**0.16**	**0.545**
Random chance	0.10	0.10	0.500

The model does not beat a single trailing window, and both sit near chance. blended_wm is flat (ρ +0.016); fqs (ρ −0.215, t −2.18) and success_pct (ρ −0.218, t −2.45) are significantly negative on their face.

The caveat is decisive — do not read this as "FQS is inverted".

With ~2 independent periods in one regime, an inverse tilt is exactly what mean reversion after a small-cap run looks like. The model's favourites were the funds that had just run hardest, and 2022-2024 punished them. That is a plausible **momentum-chasing** failure mode — a hypothesis now worth falsifying, not a conclusion.

Mechanics that did check out

- **Eligibility gate fires correctly.** At cutoff 2022-04-30 all 11 funds were excluded *insufficient_history* (required 3y, available 2.75-3.00) and no predictions were emitted.
- **Data quality is clean.** All 11 funds: 0 duplicate dates, 0 missing months, 0 long gaps, 0 detected splices, months_present == months_expected.
- **No look-ahead in stitching.** stitch_direction = backward_only (MODEL_AUDIT.md 3.1).
- **Prediction and outcome share a scale.** actual_outcome_maths = model_identical — the realised figure uses the model's own window machinery, defects included.

Comparison 5 — what was configured vs what actually ran

This matters for anyone reading the fixture parquets as "the 5-year backtest". They are not.

Setting	backtest/config.py declares	This run actually used
Rolling period	5 years	**2 years**
Validation horizon	5 years (equal by construction)	**2 years**
min_history_years	7 (5 + HISTORY_BUFFER_YEARS 2)	**3**
Universe size	full eligible universe	**11-fund fixture**
History per fund	14+ years (the target)	**~7.1 years (2019-2026)**
Independent periods	~9 non-overlapping at 5y over 2012-2026	**2.04**

The fixture was dialled down to make ~7 years of NAV produce any windows at all. results_fixture/ does not reflect current config defaults.

What would make this interpretable

The binding constraint is independent periods, and only real history buys them. On a 14-year universe at a 5-year horizon you get roughly **9 non-overlapping windows spanning 2012-2026** — covering the 2013 taper, the 2018 mid/small-cap drawdown, the 2020 crash and the 2021-24 small-cap cycle. Several regimes instead of one. That is the run which can actually test the momentum-chasing hypothesis.

#	Step	Command
1	Build the 14-year universe	`python scripts/build_long_history_universe.py`
2	Verify stitching first (4 fetches)	`... --codes 130502,140225,113177,100822 --min-years 5`
3	Re-run backtest at 5-year rolling	`set ROLLING_PERIOD_YEARS = 5, python -m backtest.run_backtest`

Generated from results_fixture (runtime 1.5s, 544 predictions, 268 validated, 17 exclusions). Figures and per-fund aggregates derived from backtest_comparison.parquet, fund_summary.parquet and accuracy_summary.json. No network data was used.